MAKILESH M

AI/ML Engineer · Data Scientist · LLMs & Voice AI Specialist

Bangalore • makilesh24225@gmail.com • +919894419452 • linkedin.com/in/makilesh/ • github.com/Makilesh/ • Portfolio

PROFESSIONAL SUMMARY

Oracle-certified AI professional with production expertise in building and deploying LLM, RAG, multi-agent frameworks and real-time voice AI with proven ability to lead technical teams and deliver AI solutions with measurable business impact.

SKILLS

AI & Machine Learning: Deep Learning, Neural Networks, NLP, Transformers, Generative AI, Large Language Models (LLMs), MLOps, Model Fine-tuning, Multi-Agent Systems, RAG, Prompt Engineering, Vector Embeddings
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, CrewAI, AutoGen, spaCy, Hugging Face, n8n, Pandas, NumPy, Matplotlib, Seaborn, NLTK, Whisper, Streamlit, Twilio
Cloud & Tools: AWS (EC2, S3, Lambda), Azure OpenAI, Docker, FastAPI, Flask, Git, SQL, MySQL, Pinecone, FAISS, ChromaDB, Weaviate, Milvus
Languages: Python, C++, C, Java, JavaScript, HTML, CSS, React.js, RESTful APIs

EXPERIENCE

AI Developer Intern | Shamla Tech Solutions | Coimbatore, India (Sep 2025 – Feb 2026)

- Deployed an **AI Voice Calling Agent** (Twilio + Pinecone) enabling fully automated outbound calls with zero manual intervention. Implemented persistent conversation memory via Pinecone vector storage, delivering context-aware dynamic responses based on prior interaction history.
- Built a full-duplex Cross-Platform Meeting Assistant with real-time speaker diarization and streaming transcription across Zoom, Google Meet, Microsoft Teams, and in-person sessions, featuring wake-word activation for on-demand contextual AI expert responses powered by RAG.
- Engineered end-to-end **Multi-source Lead Generation Pipeline** with 3-stage AI qualification (pre-filter + dual LLM: GPT-4 with Gemini, Ollama fallback), achieving 94.6% cost savings via batch processing (20 leads/API call) and concurrent async scraping across Reddit, Discord, Slack, LinkedIn with competitor detection
- **Technologies Used:** Python, LangChain, Twilio API, RealtimeSTT/TTS, Whisper, Pinecone, pyannote.audio, OpenAI / Gemini APIs, Selenium, Playwright, PRAW, Discord.py, Slack SDK, AsyncIO, Flask

PROJECTS

Agentic RAG System (Jan 2026- Feb 2026) – github.com/Makilesh/agentic_rag_system

- Engineered a multi-agent document Q&A platform using a 7-agent LangGraph state machine with a self-corrective retrieval loop that automatically rewrites low-quality queries (score < 0.7) and re-retrieves up to 2 times, improving answer accuracy across multi-format documents (PDF, DOCX, PPTX, Excel, TXT)
- Implemented true hybrid search over Milvus 2.4 vector database combining dense semantic vectors (384-dim Sentence Transformers, HNSW index) with sparse BM25-style keyword vectors via Reciprocal Rank Fusion (RRF) at ~200ms retrieval latency. Deployed via Docker Compose with Gemini 2.5 Flash primary LLM and Ollama Qwen 2.5 fallback
- **Technologies Used:** Python, LangGraph, LangChain, Milvus, Sentence Transformers, Gemini, Ollama, FastAPI, Streamlit, Docker, PyMuPDF, pandas

AI-Powered Voice Agent (Sep 2025 - Nov 2025) **-** github.com/Makilesh/voice_engine_MVP

- Engineered a production-grade full-duplex voice agent achieving 200–500ms end-to-end latency (STT + LLM + TTS) with <150ms barge-in detection via RMS energy monitoring and selective echo suppression, using a producer-consumer async architecture with Silero VAD and Whisper transcription
- Built a hybrid TTS pipeline using Kokoro-82M (40–90ms first-byte latency) with a LLM fallback chain, dynamic audio queue sizing, and memory-aware concurrency management to prevent OOM failures under sustained load.
- **Technologies Used:** Python, Asyncio, Threading, Whisper, RealtimeSTT, Kokoro, PyAudio, OpenAI, psutil, httpx

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional - 2025
- Oracle Cloud Infrastructure 2025 Certified Data Science Professional - 2025
- Udemy Complete Machine Learning, NLP Bootcamp MLOPS & Deployment - 2024
- NPTEL Python for Data Science – 2024

HACKATHONS

IMPACT X - Finalist (2024) | Team Lead, Full-Stack Developer, AI Engineer
Led a team of 4 to develop an AI-Powered Crime Detection System using YOLOv8 and CNNs.
THIRAN - Finalist (2025) | Team Lead, Data Scientist, Data Engineer
Developed AI Legal Engine with RAG architecture achieving hybrid search across 673+ legal documents

EDUCATION

Sri Krishna College of Engineering and Technology, Coimbatore, India
B.E. in Computer Science Engineering | CGPA: 8.0 | May 2026